CSE4062
Introduction to Data Science and Analytics
Spring 2023

MEMBERS

Project Details

Overview

This project involves working with a real-world dataset provided by a sponsor company in the IT support domain. The dataset consists of IT support tickets entered into a Jira system related to Oracle systems. The tickets are reported by reporters and assigned to IT specialists in the company for resolution. The dataset contains various fields such as work_log, issue_category, worker, employee_type, issue_id, Jiraname, work_log_total, work_log_ratio, and issue_sub_category, which can be used for supervised and unsupervised machine learning models.

Objective

The main objective of this project is to analyze and model the provided dataset to gain insights and develop predictive models for various tasks such as numeric prediction (regression) of work_log, classification of issue_category, worker, and employee_type.

Approach

The project will begin with exploratory data analysis (EDA) to gain insights into the dataset and understand the distribution of data in different fields. Feature engineering may be performed to extract relevant features from the dataset. Supervised machine learning models such as regression and classification algorithms may be implemented to predict work_log, issue_category, worker, and employee_type. The performance of these models will be evaluated using appropriate evaluation metrics such as accuracy, precision, recall, F1-score, and RMSE. Unsupervised machine learning models such as clustering and dimensionality reduction techniques may also be applied to gain further insights into the data.

Data Statistics

There are 40,008 rows and 18 columns in the dataset.

Target Attribute: **URGENCY**

#	Column Name	Type	Important
1	PRIORITY	Nominal	Yes
4	URGENCY	Nominal	**Target**
5	IMPACT	Nominal	Yes
7	ASSIGNEE	Text	Yes
8	POZISYON	Nominal	Yes
9	IS_BILGISI	Nominal	Yes
11	Completion Time	Numeric (float)	Yes
18	ISSUE_TYPE	Text	Yes